UrbanCart

Retail Shop Business Analytics

Q1 How many total orders has UrbanCart received so far?

SQL Query

```
select
count (distinct order_id)
from public."FactOrderItems"
```

TOTAL ORDERS

1,200

Business Insight

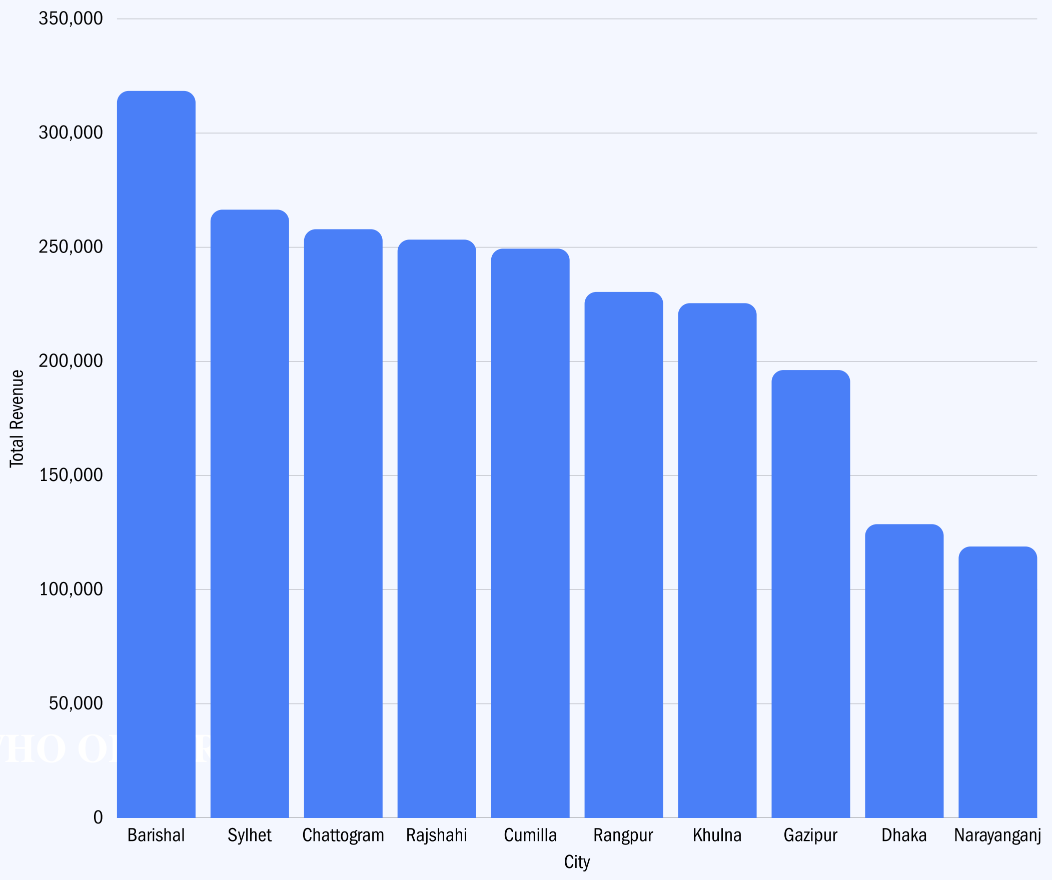
UrbanCart has processed 1,200 total orders .

Q2 Which cities generate the highest number of orders & revenue?

SQL Query

```
select
c.city as "City",
count(o.order_id) as  "Total Orders",
sum(p.quantity * q.unit_price)as "Total Revenue"
from public."FactOrders" o
join public."Dimcustomers" c on o.customer_id =
c.customer_id
join public."FactOrderItems" p on o.order_id = p.order_id
join public."DimProducts" q on p.product_id = q.product_id
group by 1
order by 2 DESC
```

city	Total Orders
Barishal	173
Sylhet	148
Chattogram	140
Rajshahi	126
Cumilla	126
Rangpur	121
Khulna	121
Gazipur	103
Dhaka	81
Narayanganj	61



Business Insight

Among all cities, Barishal got the highest number of orders (14.4%) & revenue(14.18%) of all orders.

Highest number of orders & revenue

Barishal

Q3 What percent of total customers uses gmail? ?

SQL Query

```
select
count(case when email ilike '%@gmail%' then 1
End)*100/count (Customer_id) as "Gmail Users (%)"
from public."Dimcustomers"
```

Gmail Users(%)

70%

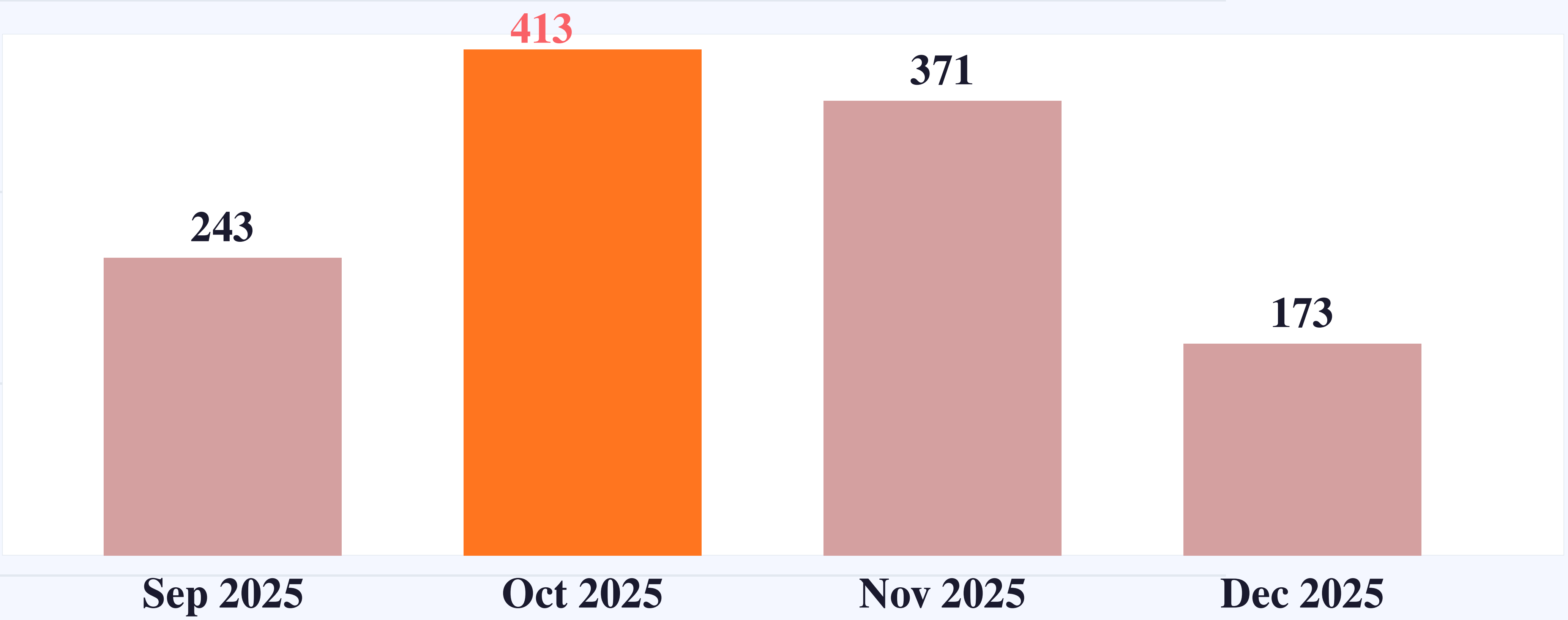
Business Insight

More than half of total customers uses gmail address.

Q4 What is the monthly trend of total orders over time?

SQL Query

```
select
To_Char(order_date:: Date, 'Month YYYY') as "Month",
Count (order_id) as "Total Orders"
from public."FactOrders"
group by 1
order by 1 desc
```



Business Insight

October 2025 was the peak month with 413 orders. December shows a sharp drop to 173.

Q5 What is the order completion rate across all orders?

SQL Query

```
Select
Round ((count(case
when status ilike 'Completed' then 1
end)*100/Count(distinct order_id)),2) as "Completion
Rate(%)"
from public."FactOrders"
```

Order Completion Rate(%)

59%

Business Insight

Only 59% of orders are completed.

Q6 What is the total revenue generated by UrbanCart?

SQL Query

```
select
sum (o.quantity*unit_price) as "Total Revenue"
from public."FactOrderItems" o
join public."DimProducts" c on o.product_id =
c.product_id
```

Total Revenue (BDT)

22,45,122

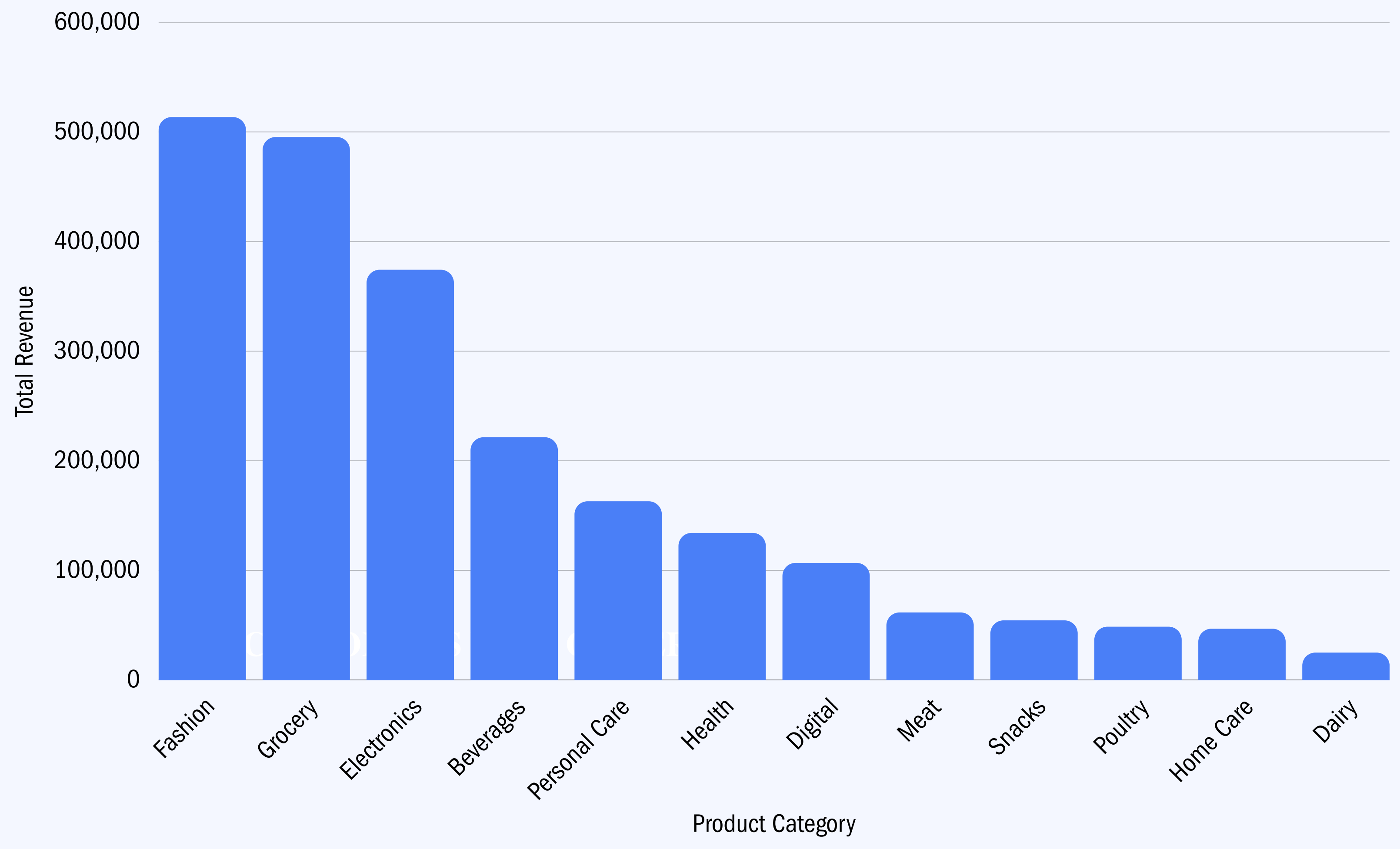
Business Insight

Total Revenue is BDT 22,45,122 .

Q7 Which product categories contribute the most to total revenue?

SQL Query

```
select
c.category as "Product Category",
sum (o.quantity*unit_price) as "Total Revenue"
from public."FactOrderItems" o
join public."DimProducts" c on o.product_id = c.product_id
group by 1
Order by 2 desc
```



Business Insight

Among all cities, Fashion got the highest revenue.

Product categories contribute the most

Fashion

Q8 Which individual products generate the highest revenue?

SQL Query

```
select
c.product_name as "Product Name",
sum (o.quantity*unit_price) as "Total Revenue"
from public."FactOrderItems" o
join public."DimProducts" c on o.product_id = c.product_id
group by 1
Order by 2 desc
```

Business Insight

Among all products, Power bank got the highest revenue.

Product categories contribute the most

Power Bank

Q9 What is the Average Order Value (AOV) and Average Basket Size?

SQL Query

```
select
round(
sum (o.quantity*c.unit_price)/count(distinct order_id),2)as "AOV"
from public."FactOrderItems" o
join public."DimProducts" c on o.product_id = c.product_id


select
round(
sum (quantity)/count(distinct order_id),2) as "Average Basket Size"
from public."FactOrderItems"
```

Average Order Value

70%

Average Basket Size

9.96

Business Insight

Result: AOV = BDT 1,870.94 | Average Basket Size = 9.96 items

Q10 Which products are at risk of stock-out due to high sales volume and low inventory?

SQL Query

```
select
product_name as "Product",
category as "Product Category",
stock as "Stock",
case
when stock <= 100 then 'Critical'
when stock <= 200 then 'Low'
when stock <= 500 then 'Medium'
else 'Healthy' end as "Status"
from public."DimProducts"
order by 3 asc
```

Product	Product Category	Stock	Status
Power Bank 10000mAh	Electronics	90	Critical
Wallet (Men)	Fashion	150	Low
Ladies Bag	Fashion	150	Low
Horlicks 500g	Health	180	Low
Broiler Chicken (whole)	Meat	200	Low
Shoes Polish	Personal Care	200	Low
Cap	Fashion	200	Low
Bru Coffee 200g	Beverages	200	Low
Clear Men Shampoo 180ml	Personal Care	250	Medium

Business Insight

Power Bank 10000mAh is the top revenue product (BDT 304,000) AND has the lowest stock (90 units) — this is a critical restocking emergency. 7 additional products are in the Low-Stock zone.

Q11 Which customers contribute the highest total revenue?

SQL Query

```
select
q.customer_id as "Customer ID",
r.full_name as "Name",
sum (o.quantity*c.unit_price) as "Total Revenue"
from public."FactOrderItems" o
join public."DimProducts" c on o.product_id = c.product_id
join public."FactOrders" q on o.order_id = q.order_id
join public."Dimcustomers" r on q.customer_id = r.customer_id

group by 1,2
Order by 3 desc
limit 10
```

SL.	Customer ID	Name	Total Revenue
1	70	Raisa	42516
2	94	Mim	38605
3	87	Sakib	38413
4	19	Arif	37013
5	96	Ehsan	36092
6	40	Siam	35673
7	29	Towhid	32653
8	88	Akash	32028
9	28	Shuvo	31983
10	77	Shahriar	31955

Business Insight

Total Revenue is BDT 22,45,122 .

Customer Details

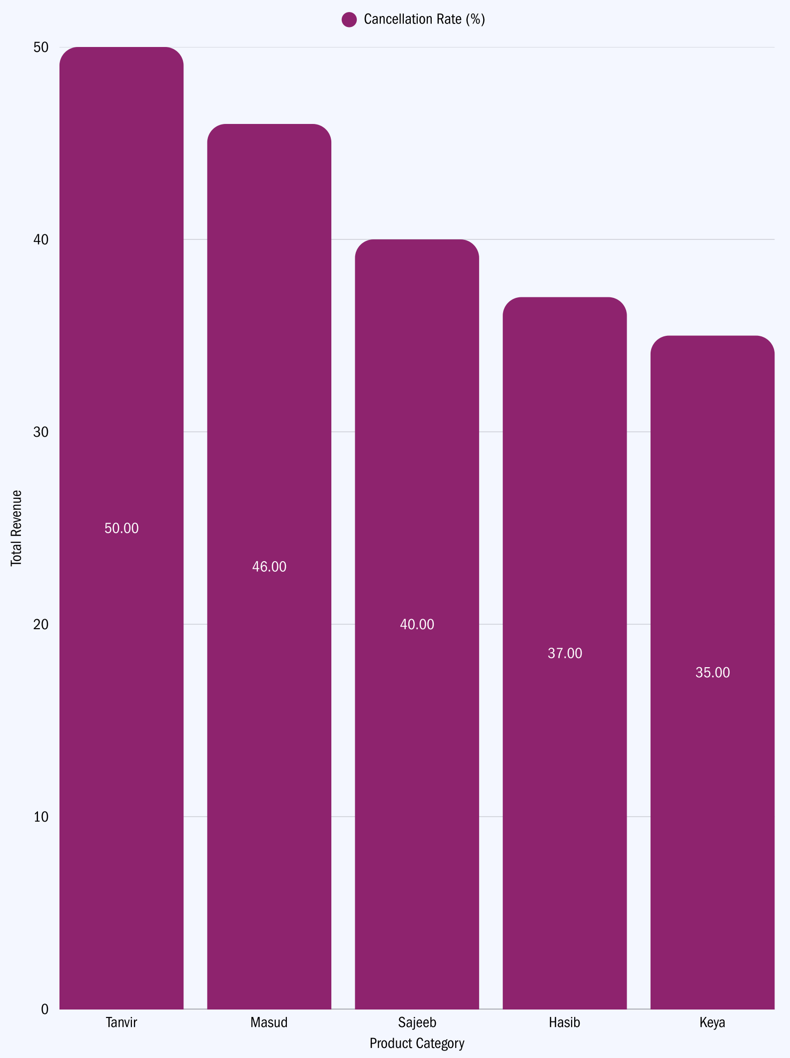
ID 70- Raisa

Q12 Generate a list of cancellation rate by city; also the cancellation rate for customers.

SQL Query

```
select
o.city as "City",
count (c.order_id) as "Total Orders",
count (case when c.status ilike '%cancelled%' then 1 end) as "Cancelled Orders",
Round(
count(case when c.status ilike '%cancelled%' then 1 end)*100/count(c.order_id),2) AS "Cancellation Rate
(%)"
from public."FactOrders" c
join public."Dimcustomers" o on c.customer_id = o.Customer_id
group by 1
order by 4 desc

select
o.full_name as "Customer Name",
count (c.order_id) as "Total Orders",
count (case when c.status ilike '%cancelled%' then 1 end) as "Cancelled Orders",
Round(
count(case when c.status ilike '%cancelled%' then 1 end)*100/count(c.order_id),2) AS "Cancellation Rate (%)"
from public."FactOrders" c
join public."Dimcustomers" o on c.customer_id = o.Customer_id
group by 1
order by 4 desc
```



Business Insight

Most cancel from Gazipur city and tanvir canceled most product.

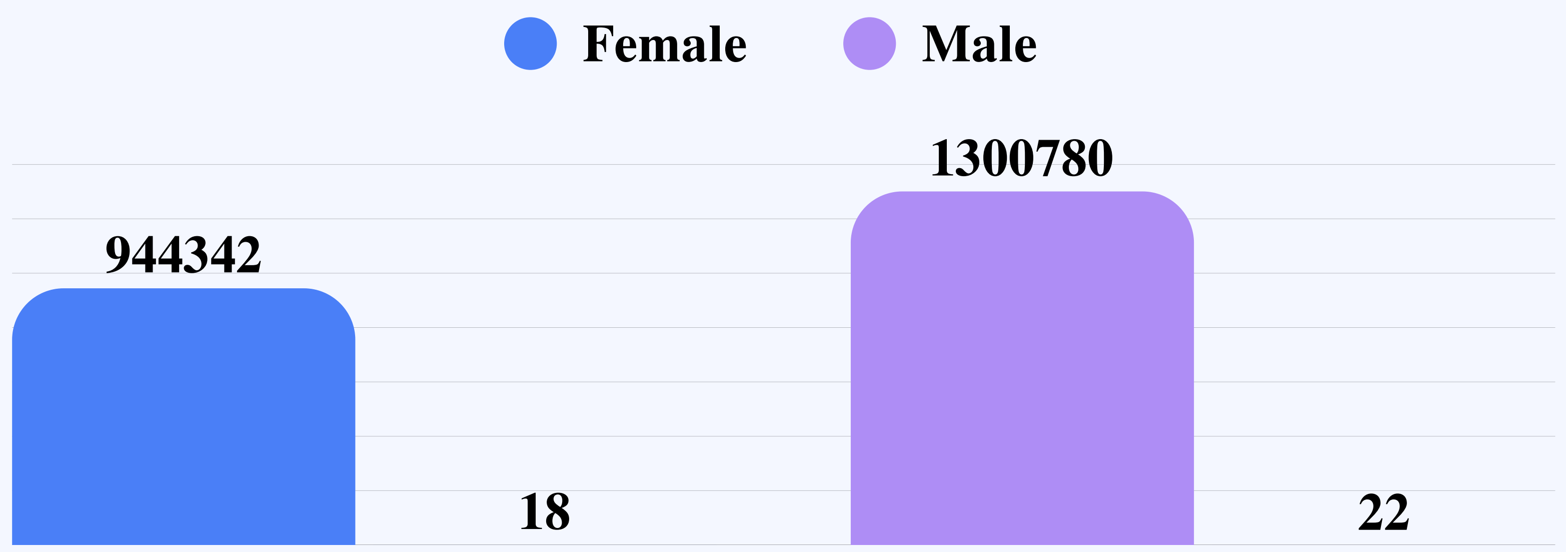
Top City & Customer in Cancellation

City-Gazipur
Customer-Tanvir

Q13 Do male and female customers show different purchasing patterns by category?

SQL Query

```
SELECT
    c."Gender" AS "Gender",
    COUNT(DISTINCT o.order_id) AS "Total Orders",
    SUM(p.unit_price * oi.quantity) AS "Total Revenue (BDT)",
    ROUND(AVG(p.unit_price * oi.quantity), 2) AS "Avg Order Value",
    COUNT(CASE WHEN o.status ILIKE '%cancelled%' THEN 1 END) AS
"Cancelled Orders",
    ROUND(COUNT(CASE WHEN o.status ILIKE '%cancelled%' THEN 1 END)
        * 100.0 / COUNT(o.order_id), 2) AS "Cancel Rate (%)"
FROM public."FactOrders" o
JOIN public."Dimcustomers" c  ON o.customer_id = c.customer_id
JOIN public."FactOrderItems" oi ON o.order_id = oi.order_id
JOIN public."DimProducts" p    ON oi.product_id = p.product_id
GROUP BY 1
```



Gender	Total Orders	Total Revenue	Avg Order Value	Cancelled Orders	Cancel Rate (%)
Female	514	944342	459.31	368	17.9
Male	686	1300780	474.05	616	22.45

Business Insight

Male customers generate 38% more revenue than Female, driven by higher order volume. However, both genders have similar cancellation rates (~30%), indicating a structural issue rather than gender-specific behavior.

Q14 How does customer purchasing behavior change over time since account creation?

SQL Query

```
SELECT
    dc.customer_id AS "Customer ID",
    dc.full_name AS "Customer Name",
CASE
 WHEN(TO_DATE(fo.order_date,'MM/DD/YYYY')  -
TO_DATE(dc.created_at,'MM/DD/YYYY')) <= 30 THEN '0-30 days'
WHEN (TO_DATE(fo.order_date,'MM/DD/YYYY') -
TO_DATE(dc.created_at,'MM/DD/YYYY')) <= 60 THEN '31-60 days'
 ELSE '60+ days'END AS "Period",
COUNT(DISTINCT fo.order_id) AS "Total Orders",
SUM(foi.quantity) AS "Total Items"

FROM public."FactOrders" fo
JOIN public."Dimcustomers" dc ON fo.customer_id = dc.customer_id
JOIN public."FactOrderItems" foi ON fo.order_id = foi.order_id

GROUP BY 1,2,3
ORDER BY 1,2,3
```

Customer ID	Customer Name	Period	Total Orders	Total Items
1	Abdullah	0-30 days	5	49
1	Abdullah	31-60 days	2	18
1	Abdullah	60+ days	1	7
2	Ayon	0-30 days	3	27
2	Ayon	31-60 days	5	47
2	Ayon	60+ days	1	10

Business Insight

Customer order ratios vary across time periods since account creation.

Q15 Generate a list of customers who ordered in October but didn't order in December.

SQL Query

```
SELECT DISTINCT
    c.customer_id as "Customer ID",
    c.full_name AS "Customer Name",
    c.city as "Customer City"
FROM public."Dimcustomers" c
WHERE c.customer_id IN (
    SELECT customer_id FROM public."FactOrders"
    WHERE EXTRACT(MONTH FROM TO_DATE(order_date,
'MM/DD/YYYY')) = 10)
AND c.customer_id NOT IN (
    SELECT customer_id FROM public."FactOrders"
    WHERE EXTRACT(MONTH FROM TO_DATE(order_date,
'MM/DD/YYYY')) = 12)
```

SL.	Customer ID	Customer Name
1	2	Ayon
2	5	Samia
3	12	Lamya
4	14	Kamal
5	15	Hasib
6	16	Heba
7	17	Zannat
8	21	Mahjabin
9	22	Ehsan

SL.	Customer ID	Customer Name
10	23	Tanvir
11	26	Riya
12	36	Rakib
13	47	Sumaiya
14	65	Afrin
15	66	Rodela
16	84	Farzana
17	99	Mahmud

Business Insight

Total 17 customers ordered in October but didn't order in December.

Q16 Generate a list of customers who ordered both in October & December.

SQL Query

```
SELECT DISTINCT
 c.customer_id as "Customer ID",
 c.full_name AS "Customer Name",
 c.city as "Customer City"
FROM public."Dimcustomers" c
WHERE c.customer_id IN (
    SELECT customer_id FROM public."FactOrders"
    WHERE EXTRACT(MONTH FROM TO_DATE(order_date,
'MM/DD/YYYY')) = 10)
AND c.customer_id  IN (
    SELECT customer_id FROM public."FactOrders"
    WHERE EXTRACT(MONTH FROM TO_DATE(order_date,
'MM/DD/YYYY')) = 12)
```

Customer ID	Customer Name	Customer City
1	Abdullah	Chattogram
3	Taushif	Khulna
4	Saiful	Rangpur
6	Shithi	Chattogram
7	Rumana	Chattogram
8	Lamisa	Sylhet
9	Anika	Cumilla
10	Irtaza	Gazipur

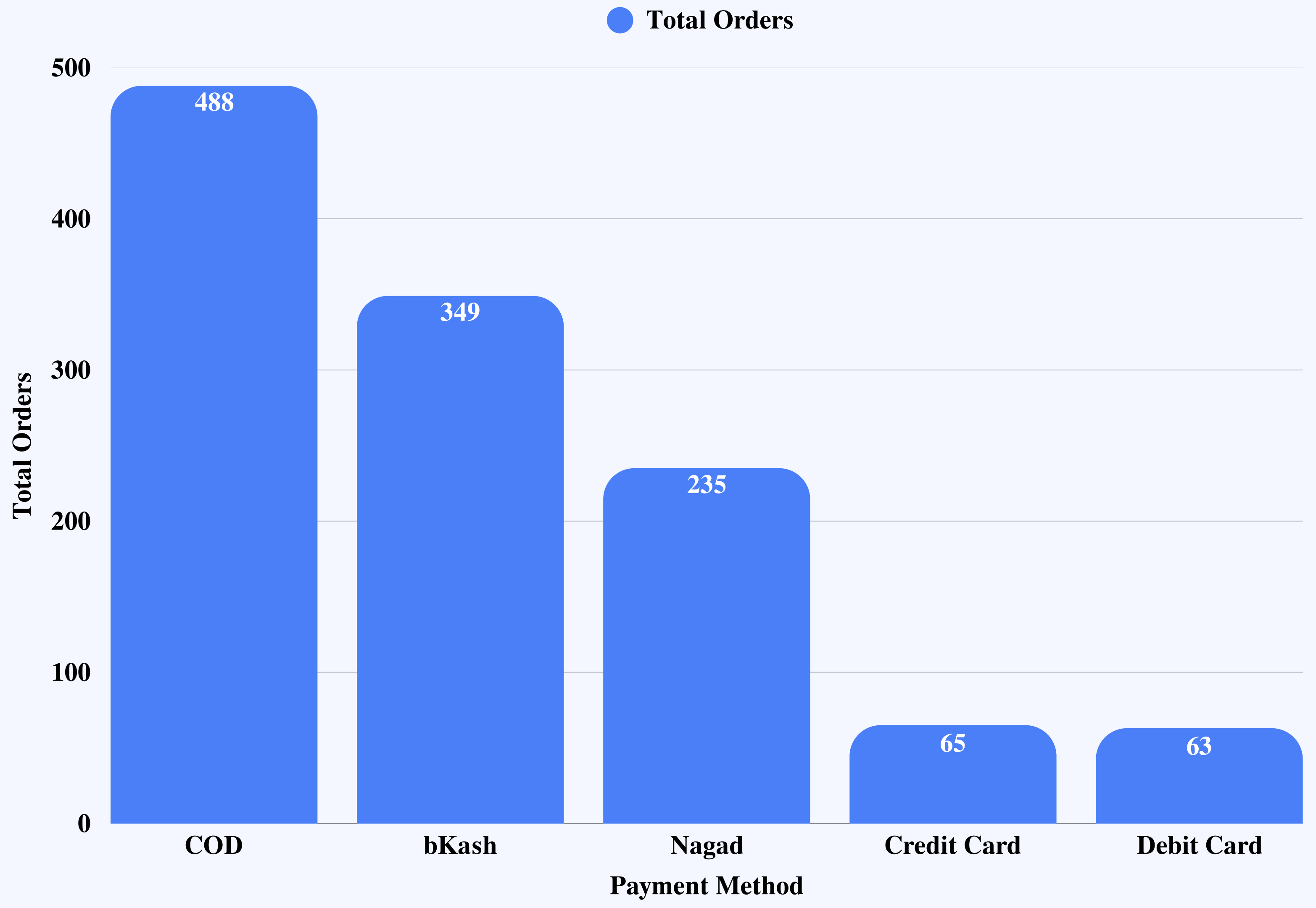
Business Insight

Total 81 customers ordered in October & December.

Q17 Which payment methods are used most frequently?

SQL Query

```
select
method as "Payment Method",
count (Distinct "order_id")as "Total Orders"
from public."FactPayment"
group by 1
order by 2 desc
```



Business Insight

Cash on Delivery method most customer prefered.

Top Method

Cash on Delivery

Q18 Is there any relationship between payment method and order status?

SQL Query

```
select
o.method as "Payment Method",
c.status as "Order Status",
count (Distinct o."order_id")as "Total Orders"
from public."FactPayment" o
join public."FactOrders" c on o.order_id =
c.order_id
group by 1,2
order by 1,3 desc
```

Payment Method	Order Status	Total Orders
bKash	Completed	202
bKash	Pending	74
bKash	Cancelled	73
COD	Completed	300
COD	Cancelled	97
COD	Pending	91
Credit Card	Completed	39
Credit Card	Pending	15
Credit Card	Cancelled	11
Debit Card	Completed	35
Debit Card	Cancelled	18
Debit Card	Pending	10
Nagad	Completed	137
Nagad	Pending	51
Nagad	Cancelled	47

Business Insight

COD has highest completions (300). Debit Card has highest cancellation ratio (~29%).

Q19 Do certain cities prefer specific payment methods?

SQL Query

```
select
p.city as "City",
o.method as "Payment Method",
count (Distinct o."payment_id")as "Transaction Count",
Round (count (Distinct
o."payment_id")*100/sum(Count(o."payment_id"))
over (partition by p.city,2)) as "City Share (%)"
from public."FactPayment" o
join public."FactOrders" c on o.order_id = c.order_id
join public."Dimcustomers" p on c. customer_id = p.customer_id
group by 1,2
order by 1,3 desc
```

City	Payment Method	Transaction Count	City Share (%)
Barishal	COD	76	44
Barishal	bKash	44	25
Barishal	Nagad	28	16
Barishal	Debit Card	15	9
Barishal	Credit Card	10	6
Chattogram	COD	56	40
Chattogram	bKash	44	31
Chattogram	Nagad	34	24
Chattogram	Credit Card	4	3
Chattogram	Debit Card	2	1

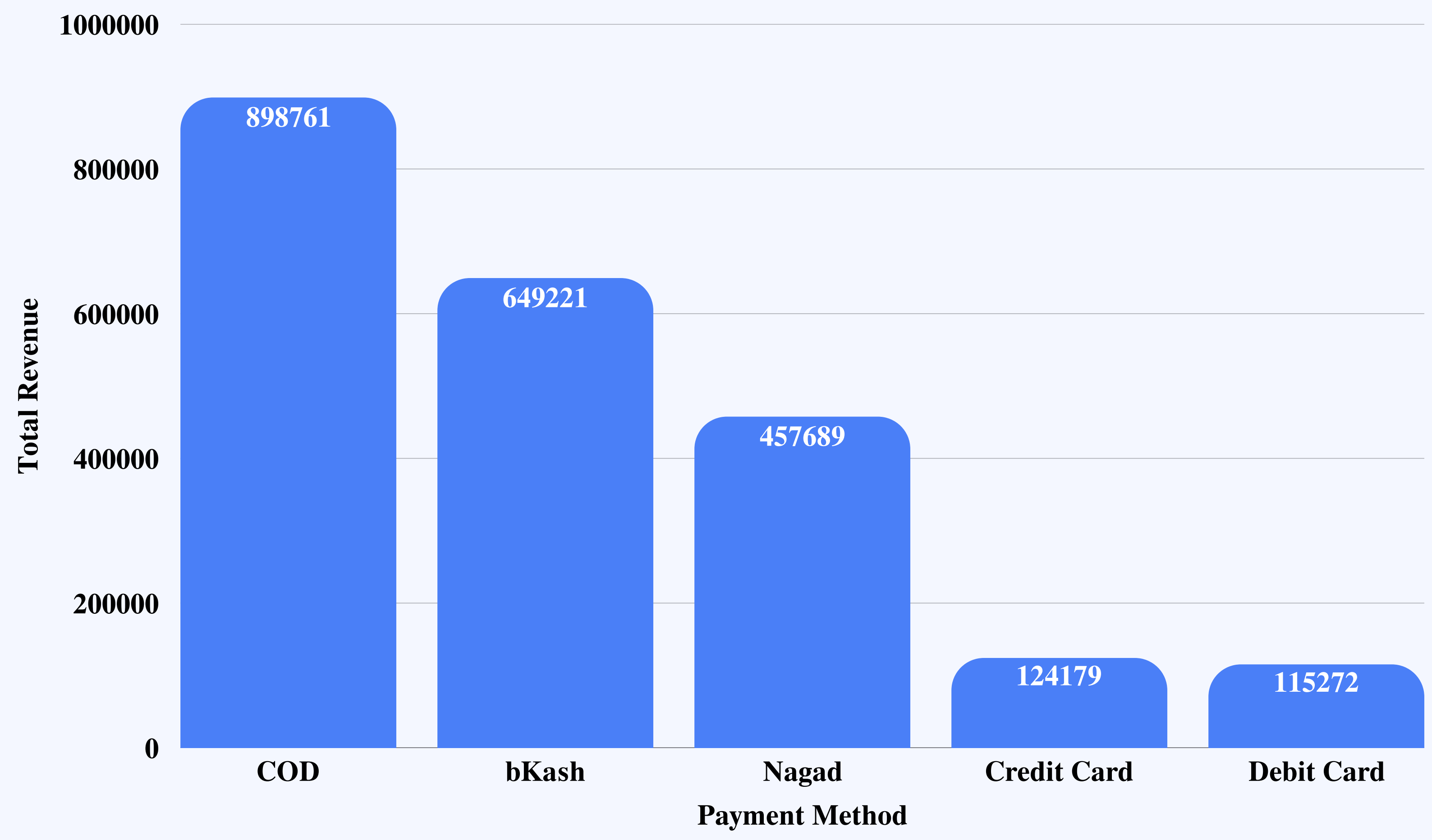
Business Insight

All cities prefer Cash on Delivery payment methods.

Q20 Are higher-value orders associated with specific payment methods?

SQL Query

```
select
o.method as "Payment Method",
sum(p.quantity * q.unit_price)as "Total Revenue"
from public."FactPayment" o
join public."FactOrderItems" p on o.order_id =
p.order_id
join public."DimProducts" q on p.product_id =
q.product_id
group by 1
order by 2 DESC
```



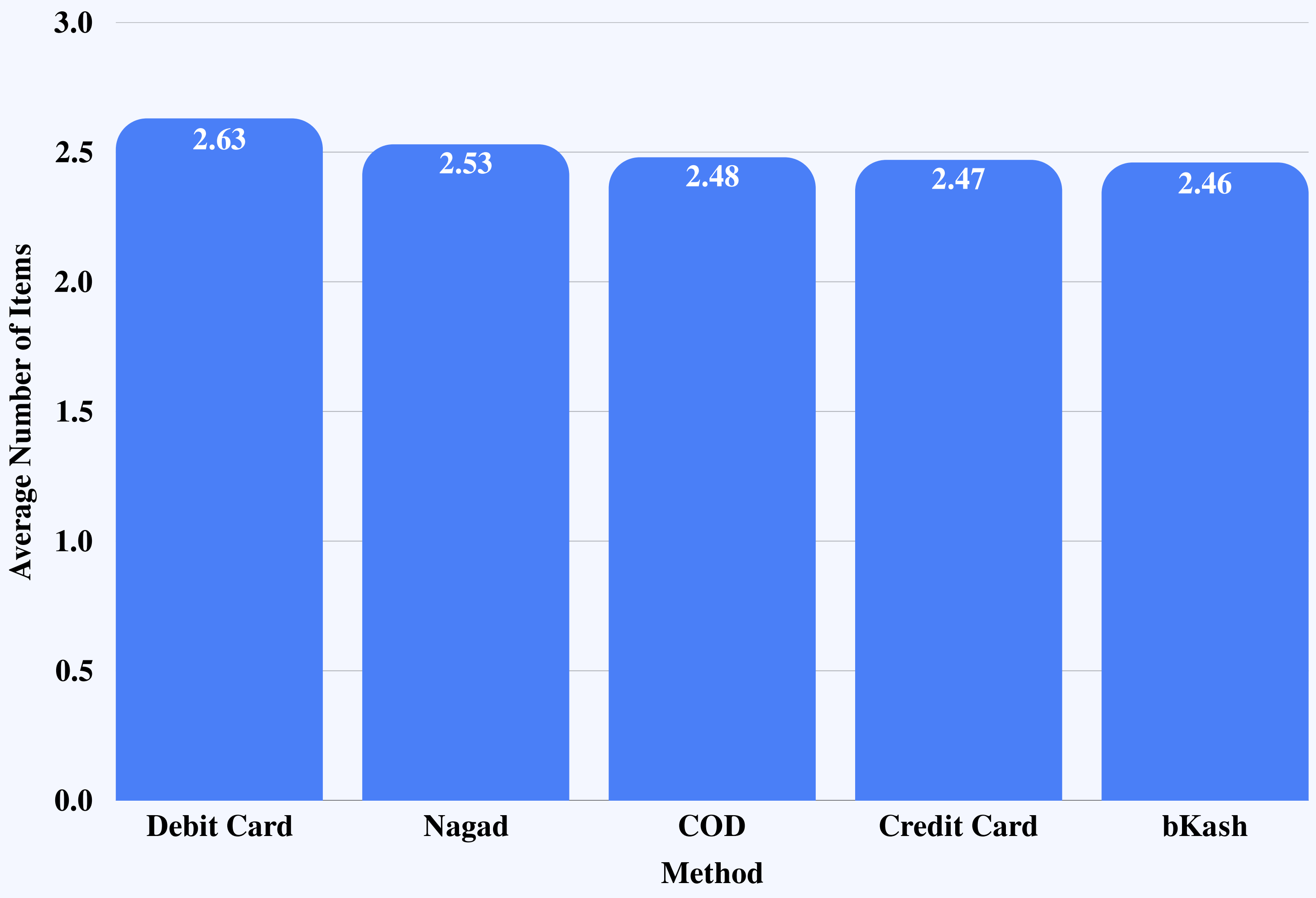
Business Insight

Higher-value orders associated with Cash on Delivery Method.

Q21 What is the average number of items per order by payment method?

SQL Query

```
select
c.method as "Method",
round (Avg (o.quantity),2) as "Average Number of
Items"
from public."FactOrderItems" o
join public."FactPayment" c on o.order_id =
c.order_id
group by 1
order by 2 desc
```



Business Insight

Highest average number of items per order paid by Debit Card payment method.

Q22 Create a list to show the difference between the category average price andproduct's price for all the products.

SQL Query

```
select
product_name as "Product Name",
category as "Product Category",
Unit_price as "Unit Price",
Round(avg(unit_price) over(partition by category),2) as
"Category Average Price",
unit_price - Round(avg(unit_price) over(partition by
category),2) as "Difference"
from public."DimProducts"
order by 5 asc
```

Product Name	Product Category	Unit Price	Category Average Price	Difference
Earphones	Electronics	250	600	-350
Cap	Fashion	120	380	-260
Oral Saline (ORS)	Health	12	246	-234
Green Chili 100g	Grocery	20	164.3	-144.3
Potato 1kg	Grocery	35	164.3	-129.3
ACI Pure Salt 1kg	Grocery	38	164.3	-126.3
Water Bottle 1L	Beverages	20	137	-117

Business Insight

The highest difference between the category average price andproduct's price for Earrphones.

Q23 Which product pairs generate the highest total revenue when sold together?

SQL Query

```
SELECT
   f1.product_id AS "Product 1",
   f2.product_id AS "Product 2",
   COUNT(*) AS "Pair Count"
FROM public."FactOrderItems" f1
JOIN public."FactOrderItems" f2
   ON f1.order_id = f2.order_id
   AND f1.product_id < f2.product_id
GROUP BY 1,2
ORDER BY 3 DESC;
```

Product 1	Product 2	Pair Count
24	30	22
8	24	21
23	34	18
27	37	18
4	33	18
1	40	18
7	22	17
9	40	17
28	33	17

Business Insight

Product 24 & 30 Pairs ordered highest time.

Top Paired

Product 24 & 30

Q24 Which product pairs generate the highest total revenue when sold together?

SQL Query

```
SELECT
    f1.product_id AS "Product 1",
    f2.product_id AS "Product 2",
    SUM(
        (p1.quantity * p2.unit_price) +(p1.quantity * p2.unit_price)) AS "Total Revenue"

FROM public."FactOrderItems" f1
JOIN public."FactOrderItems" f2
    ON f1.order_id = f2.order_id
    AND f1.product_id < f2.product_id

JOIN public."FactOrderItems" p1
    ON f1.product_id = p1.product_id

JOIN public."DimProducts" p2
    ON f2.product_id = p2.product_id

GROUP BY 1,2
ORDER BY 3 DESC;
```

Product 1	Product 2	Total Revenue
9	40	11369600
1	40	10054800
15	40	8481600
24	40	8299200
33	40	7934400

Business Insight

Product 9& 40 Pairs generate the highest revenue.

Top Paired

Product 09 & 40

Q25 Generate a daily report containing the columns Date, Total Orders, Total Items, Completed Orders, Cancelled Orders, Total Revenue

SQL Query

```
SELECT
    TO_DATE(o.order_date, 'MM/DD/YYYY') AS "Order Date",
COUNT(DISTINCT o.order_id) AS "Total Orders",
SUM(c.quantity) AS "Total Items",

COUNT(DISTINCT CASE
        WHEN o.status = 'Completed' THEN o.order_id END) as "Completed Orders",
COUNT(DISTINCT CASE
          WHEN o.status = 'Cancelled' THEN o.order_id END) as "Cancelled Orders ",
Count(Distinct Case
when o.status='Pending' then o.order_id end) as "Pending Orders)",

SUM(c.quantity * q.unit_price) as "Total Revenue"
FROM public."FactOrders" o
JOIN public."FactOrderItems" c on o.order_id = c.order_id
JOIN public."DimProducts" q on c.product_id = q.product_id
GROUP BY 1
ORDER BY 1
```

Order	Total	Total	Complete	Cancelled	Pending	Total
9/14/2025	15	154	10	0	5	24172
9/15/2025	11	105	7	1	3	16110
9/16/2025	14	135	8	3	3	29315
9/17/2025	21	201	14	2	5	32249
9/18/2025	8	73	3	1	4	11946
9/19/2025	6	60	6	0	0	10393
9/20/2025	13	124	8	2	3	20656
9/21/2025	13	131	10	1	2	24629
9/22/2025	12	129	8	4	0	27080
9/23/2025	20	198	13	2	5	38096

Business Insight

Daily breakdown helps identify peak dates and fulfillment patterns over time.